Roads Are No Fun

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Final / Critical Design Review

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1 Introduction

Design reviews serve as gating events that establish the progress attained during the design process for a product and determine whether the next steps of design/production can begin, or if additional work is required. The final or critical design review is the review which evaluates if the design is complete and ready for production. Obviously at this stage of design, all requirements and constraints must be expected to be met, or have a clear path identified to be met.

2 Expectations

2.1 Front Matter

point of contact / author / organization
- date, revision
- subject / identifiers / part numbers
- contents

2.2 Design Overview

context
- purpose
- function
- overall design concept / architecture
- consists of

2.3 Design Criteria

guidance documents / standards
- requirements:
- performance criteria
- functional criteria
- requirements verification matrix
- key performance parameters
- higher level driving requirements
- derived requirements
- concept specific requirements

2.4 Design Details

detailed architecture
- what and how
- standard / COTS components
- materials
- trade studies / alternatives / evaluations

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2.5 Lifecycle Summary

manufacture
- processes
- assembly
- operation
- maintenance
- disposal
- tooling and test equipment
- facilities

2.6 Design Verification

verification plan: inspection, analysis, test, demonstrate, similarity, lot testing
- qualification and acceptance
- Analysis -methodology -margins summary -loads and environments -analysis results
- durability: fracture / fatigue / wear
- Testing: test plan

2.7 Safety & Failure

FMECA
- Hazards
- Hazard controls
- risks and watch items

2.8 Forward Work

Identify and remaining open items
- schedule and cost estimates

2.9 Success / Exit Criteria

design is complete
- design meets all requirements and constraints
- feasible to build
- ready for implementation / production

References

1. MIL-A-83577, 1988.

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